Supplement to

Plants in limbo — the theory of detrital photosynthesis

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Supplementary methods

Halophila ovalis ramets with several leaf pairs and intact roots were collected from Pelican Point in the brackish Swan Estuary (31.98715°S, 115.82101°E) alongside sediment from adjacent Matilda Bay (31.97572°S, 115.82324°E) on 15th September 2022. I filled the bottom of a 220-L glass flow-through seawater tank with the sediment and planted the ramets to acclimatise them to ambient seawater temperature and salinity and a 12-12-h light-dark cycle (11 ± 1.6 µmol photons m^{-2} s^{-1}, mean ± standard deviation, MQ-510, Apogee Instruments Inc., Logan, USA, cf. Masini & Manning 1997) for about two hours. *Amphibolis antarctica* stems with several leaf clusters were collected from fully marine Marmion Lagoon (31.85303°S, 115.75092°E) on 13th October 2022 and immediately used in the experiment. Sediment and seagrasses were collected from less than one metre depth.

The ex situ decomposition experiment was commenced on the respective day of collection by excising individual leaves (*H. ovalis*) or leaf clusters (*A. antarctica*), hereinafter leaves. Leaves were separated from the rest of the plant by cutting the petiole (*H. ovalis*) or stem (*A. antarctica*) just below the leaf base. Excision prematurely induces senescence and was intended to mimic detachment of non-senescent leaves, which may be caused by hydrodynamics or herbivory, rather than abscission. Photosynthesis of six (*H. ovalis*) or nine (*A. antarctica*) samples was immediately measured (see next paragraph). The remaining leaves were enclosed in mesh (3-mm diameter). Mesh bags were weighed down on the sediment with inert aquarium pebbles and kept submerged in the abovementioned 220-L tank for the duration of the experiment. I maintained the 12-h light phase at 11 ± 1.6 µmol photons m^{-2} s^{-1} because this irradiance is sufficient for detrital photosynthesis to play a role (Wright et al. 2024) and is representative of the light environment experienced by detritus in the understorey of kelp forests and seagrass meadows or on deeper sediment flats. To measure detrital photosynthesis, six destructive leaf samples were taken at each of five (*A. antarctica*) to six (*H. ovalis*) timepoints over 34 days following excision.

Light-saturated net photosynthesis (P_{max}) of leaves was measured using closed oxygen (O_2) incubations (cf. Wright et al. 2024). *A. antarctica* leaves weigh roughly ten times as much as *H. ovalis* leaves (given my data the mass ratio seems to be closer to 14:1). Since I wanted to measure both seagrasses at a comparable leaf mass per volume for the same duration, I incubated ten *H. ovalis* leaves or single *A. antarctica* leaves alongside seawater blanks in sealed glass jars filled with water collected from the flow-through system. Each jar was equipped with a magnetic stir bar and a self-adhesive planar O_2 sensor spot (SP-PSt3-SA-NAU-D5-YOP, PreSens Precision Sensing GmbH, Regensburg, Germany) and placed on a magnetic stirrer under a saturating (Masini & Manning 1997, Jamaludin et al. 2006, Said et al. 2021) irradiance of 420 ± 19 µmol photons m^{-2} s^{-1} (Zeus, Ledzeal, Shenzhen Topline Lighting Technology Co. Ltd., Shenzhen, China) in a 20°C room. Care was taken to exclude air from jars. Dissolved O_2 (µM) was measured fibre-optically through the glass every 10 s over at least 35 min with a four-channel O_2 meter (OXY-4 SMA G2, PreSens Precision Sensing GmbH, Regensburg, Germany). The O_2 meter was calibrated using anoxic (1% w/v Na_2SO_3) and air-saturated (bubbled with air) ultrapure water. All measurements were corrected for incubation temperature (18 ± 0.73 °C), pressure (1020 ± 6 hPa), and salinity (35 ± 0.48 ‰), respectively using a single temperature dipping probe connected to the O_2 meter and placed in a jar filled with seawater, the built-in pressure sensor of the O_2 meter, and a handheld refractometer. I blotted and weighed (0.01-g accuracy, 440-33N, Kern & Sohn GmbH, Balingen, Germany) leaves and gravimetrically determined jar volume with ultrapure water to standardise P_{max} as shown in Equation S1.

In the first instance, simple linear models with centred incubation time ($t - \bar{t}$, min) as the predictor and dissolved O_2 (μM) as the response variable were fit to measurements from each sample and blank incubation. This yielded posterior probability distributions for slopes (β, μM min^{-1}) which were converted to mass-based P_{max} (μmol g^{-1} h^{-1}) as

$$P_{max} = \frac{(\beta_s - \beta_b) \times V \times \Delta t}{m} \quad \text{(S1)}$$

where subscript s and b denote sample and blank incubations from the same measurement group, V is the incubation volume (175 ± 0.17 mL) in litres, m is the sample blotted mass (0.52 ± 0.23 g) in grams and Δt is the desired period (1 h) in minutes. I also calculated leaf-based P_{max} because the leaves of *H. ovalis* and leaf clusters *A. antarctica* are quite uniform in size which may lead to less variation in leaf- than mass-based P_{max}. For *A. antarctica*, leaf-based P_{max} was simply $(\beta_s - \beta_b) \times V \times \Delta t$. For *H. ovalis*, the rate was additionally divided by 10 because there were 10 leaves in each incubation chamber.

Varying conditions during the 35-min incubations described above could affect P_{max} and confound the experimental effect. Possible confounding variables that I measured are initial incubation O_2 (O, μM), mean incubation temperature (T, °C), mean incubation pressure (P, hPA), incubation salinity (S, ‰) and sample mass (m, g). Although the oxygen measurements themselves were corrected for temperature, pressure and salinity, photosynthesis could also be influenced by these variables. Similarly, even though P_{max} was standardised by mass (Equation S1), the amount of leaf relative to water in the incubation chamber could additionally affect photosynthesis by changing the availability of carbon and nutrients. I explored these unwanted effects using a multiple linear regression of P_{max} against standardised (minus mean, divided by standard deviation) confounders:

$$\begin{aligned}
P_{max_i} &\sim \text{Normal}(\ \overline{P_{max_i}}\ , \sigma_{P_{max}}\ ) \quad \text{(S2)}\\
\overline{P_{max_i}} &= \alpha + \beta_O O_i + \beta_T T_i + \beta_P P_i + \beta_S S_i + \beta_m m_i\\
\alpha &\sim \text{Normal}(\ 21\ ,\ 10\ )\\
\beta_O, \beta_T, \beta_P, \beta_S, \beta_m &\sim \text{Normal}(\ 0\ ,\ 1\ )\\
\sigma_{P_{max}} &\sim \text{Exponential}(\ 1\ )
\end{aligned}$$

The likelihood for P_{max} is normal because predictions of net photosynthesis can take negative values. Each parameter was stratified by species. Even though P_{max} was only weakly associated with confounders (Figure S1), I decided to account for them in the main model. To predict P_{max} with detrital age (t, days), I used the logistic model of detrital photosynthesis (Equation 1) as

$$\begin{aligned}
P_{max_i} &\sim \text{Normal}(\ \overline{P_{max_i}}\ , \sigma_{P_{max}}\ ) \quad \text{(S3)}\\
\overline{P_{max_i}} &= \frac{\alpha_i + \tau_i}{\frac{1 + e^{\frac{5(t_i - \mu_i)}{\mu_i}}}{}} - \tau_i\\
\ln \alpha_i &= \overline{\ln \alpha}\ + \beta_O O_i + \beta_T T_i + \beta_P P_i + \beta_S S_i + \beta_m m_i\\
\ln \mu_i &= \overline{\ln \mu}\ + \gamma_O O_i + \gamma_T T_i + \gamma_P P_i + \gamma_S S_i + \gamma_m m_i\\
\ln \tau_i &= \overline{\ln \tau}\ + \delta_O O_i + \delta_T T_i + \delta_P P_i + \delta_S S_i + \delta_m m_i\\
\overline{\ln \alpha} &\sim \text{Normal}(\ \ln 21\ ,\ 0.4\ )\\
\overline{\ln \mu} &\sim \text{Normal}(\ \ln 14\ ,\ 0.6\ )\\
\overline{\ln \tau} &\sim \text{Normal}(\ \ln 2\ ,\ 0.4\ )\\
\beta_O, \beta_T, \beta_P, \beta_S, \beta_m, \gamma_O, \gamma_T, \gamma_P, \gamma_S, \gamma_m, \delta_O, \delta_T, \delta_P, \delta_S, \delta_m &\sim \text{Normal}(\ 0\ ,\ 1\ )\\
\sigma_{P_{max}} &\sim \text{Exponential}(\ 1\ )
\end{aligned}$$

where α is baseline photosynthesis (μmol g^{-1} h^{-1}), μ is the photosynthetic half-life (days) and τ is respiration of dead leaves (μmol g^{-1} h^{-1}). These parameters are strictly positive, so their logarithm is described by a multiple linear regression against the confounders (cf. Equation S2). Some species-specific influence of confounders on α and μ is evident (Figure S1), so their inclusion in this model is important for accurate prediction. I also ran this model for leaf-based P_{max}, adjusting the priors for α and τ accordingly.

Prior probability distributions for parameters were chosen based on published estimates. For the initial linear regressions of O_2 against incubation time, I centred the intercept prior on regional average seawater O_2 and the slope prior on zero. For α (Equation S2, S3), I chose the prior mean based on various published P_{max} estimates for *H. ovalis* (40 ± 57 µmol g^{-1} h^{-1}, 1.8 ± 2.5 µmol $leaf^{-1}$ h^{-1}, n = 32, Björk et al. 1997, Jamaludin et al. 2006, Borum et al. 2016, Lamit & Tanaka 2021, Said et al. 2021, 2026) and A. *antarctica* (14 ± 9 µmol g^{-1} h^{-1}, 8.8 ± 5.5 µmol $leaf^{-1}$ h^{-1}, n = 8, Masini & Manning 1997, Borum et al. 2016, Said et al. 2026). Units were converted where necessary by multiplying by mean dry-fresh mass ratios for *H. ovalis* (0.21 g dry mass g^{-1} fresh mass) and A. *antarctica* (0.29 g dry mass g^{-1} fresh mass) (de los Santos et al. 2012, Borum et al. 2016) and my own leaf fresh masses for *H. ovalis* (0.04 ± 0.01 g $leaf^{-1}$) and A. *antarctica* (0.61 ± 0.27 g $leaf^{-1}$). For μ (Equation S3), I settled on a median of 14 days, close to the minimum of 15 days for *Ecklonia radiata* (Wright et al. 2024) because I expected photosynthetic decay to be faster than for seaweeds. For τ (Equation S3), I centred the prior around published estimates of O_2 consumption by leaf litter of *Zostera marina* (17 ± 5.6 µmol g^{-1} h^{-1}, n = 16, Blum & Mills 1991) and *Posidonia oceanica* (1.3 ± 1.3 µmol g^{-1} h^{-1}, n = 46, Mateo & Romero 1996). Units were again converted by multiplying by mean dry-fresh mass ratios for *Z. marina* (0.23 g dry mass g^{-1} fresh mass, Evans et al. 1986) and *P. oceanica* (0.24 g dry mass g^{-1} fresh mass, Apostolaki et al. 2024). I chose a reasonable standard deviation for each prior based on prior simulation and variation in the published estimates.

The treatment of measurement error deserves brief mention. β and V (Equation S1) and hence P_{max} (Equation S1, S2, S3) as well as O and T (Equation S2, S3) are measured with error, which I propagated into each downstream model as the standard deviation of upstream posteriors. This was achieved by parameterising latent variables for P_{max}, O and T. In all cases I used the error structure mean ~ Normal(latent , standard deviation) where each observed mean is described by a normal distribution centred on the unobserved true value with observed error (McElreath 2020). For the response variable P_{max}, the latent variable is described by the normal likelihood (Equation S2, S3). For the standardised predictor variables O and T, the latent variable is described by a standard normal prior: latent ~ Normal(0 , 1). Wherever a variable is measured with error, I visualised each observation as a univariate or bivariate distribution (violin or ellipse) rather than a point.

Data analysis and visualisation were performed in R v4.5.1 (R Core Team 2026) within the integrated development environment RStudio v2026.01.0+392 (RStudio Team 2026). General workflow was facilitated by the tidyverse package family v2.0.0 (Wickham et al. 2019) and here v1.0.2 (Müller 2025). Hamiltonian Monte Carlo models were written in Stan (Carpenter et al. 2017) and run with CmdStan v2.17.1 (Lee et al. 2026) via the R interface cmdstanr v0.9.0 (Gabry et al. 2025). All models were run with 8 Markov chains spread across all cores with 10^4 warmup and sampling iterations each. Convergence and smooth sampling were optimised by assessing $\hat{R}$ scores and visually scrutinising trace rank and pair plots with bayesplot v1.14.0 (Gabry et al. 2019). All reported results are posteriors and derived central tendencies and intervals calculated with tidybayes v3.0.7 (Kay 2024). Figure and table creation was facilitated by ggridges v0.5.7 (Wilke 2021), ggh4x v0.3.1 (van den Brand 2025), geomtextpath v0.2.0 (Cameron & van den Brand 2022), patchwork v1.3.2 (Pedersen 2025) and officer v0.7.0 (Gohel et al. 2025). For details, all data, code, diagnostic plots and priors and posteriors can be consulted at github.com/lukaseamus/plant-limbo and doi.org/10.5281/zenodo.23182864. Illustrations were made with Affinity Designer v1.10.8 (Serif Ltd., Nottingham, UK).

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Supplementary figures

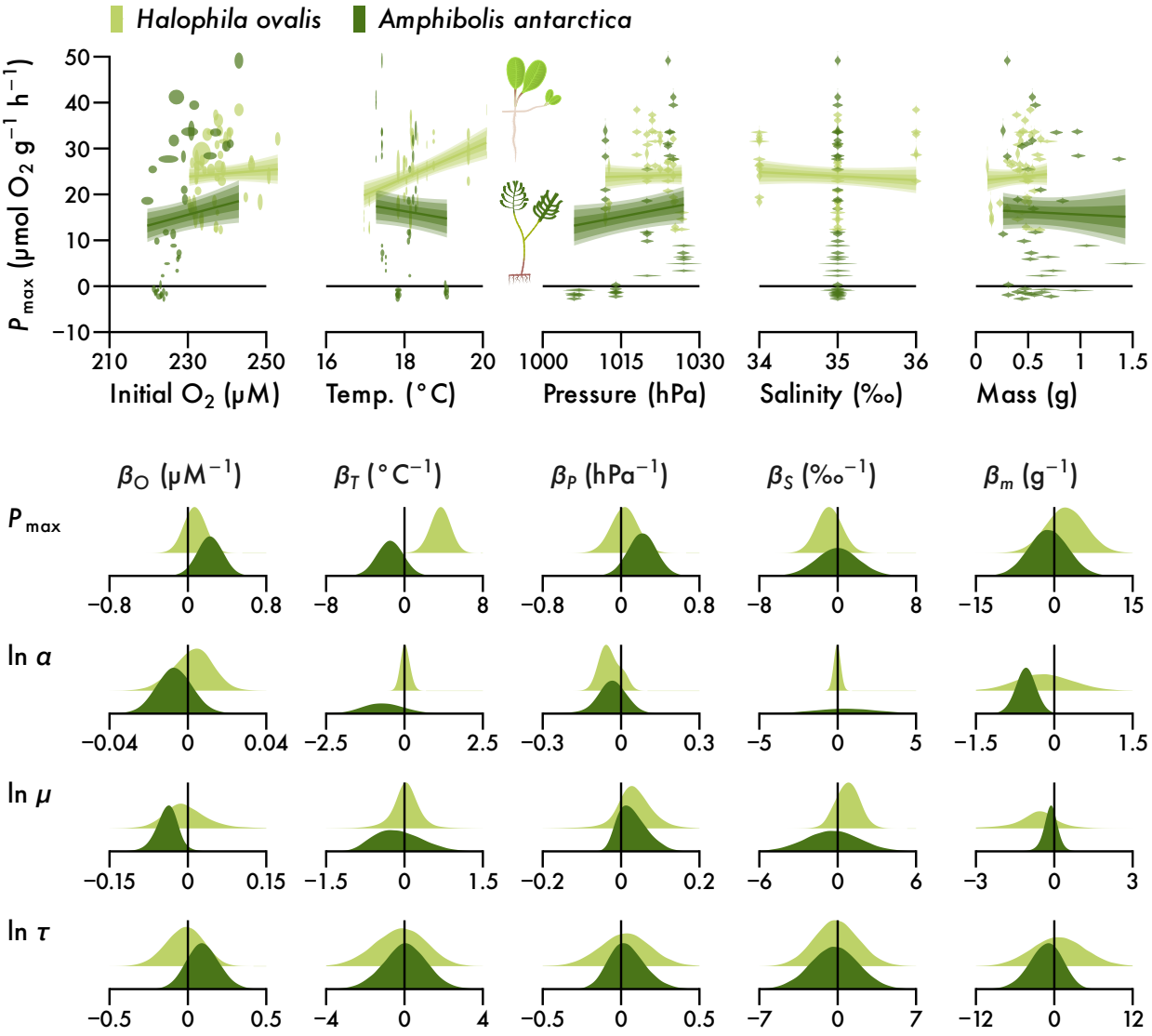


Figure S1. Effect of confounding variables on light-saturated net photosynthesis (P_{max}) of excised *Halophila ovalis* and *Amphibolis antarctica* leaves. P_{max} is standardised by leaf fresh mass. Ellipses and violins are kernel density estimates of observations with measurement error (n = 42 for *H. ovalis*, n = 39 for *A. antarctica*). Lines and intervals are medians and the central 50%, 80% and 90% of posterior probability for predicted means. Distributions are kernel density estimates of slope posteriors (Equation S2, S3).

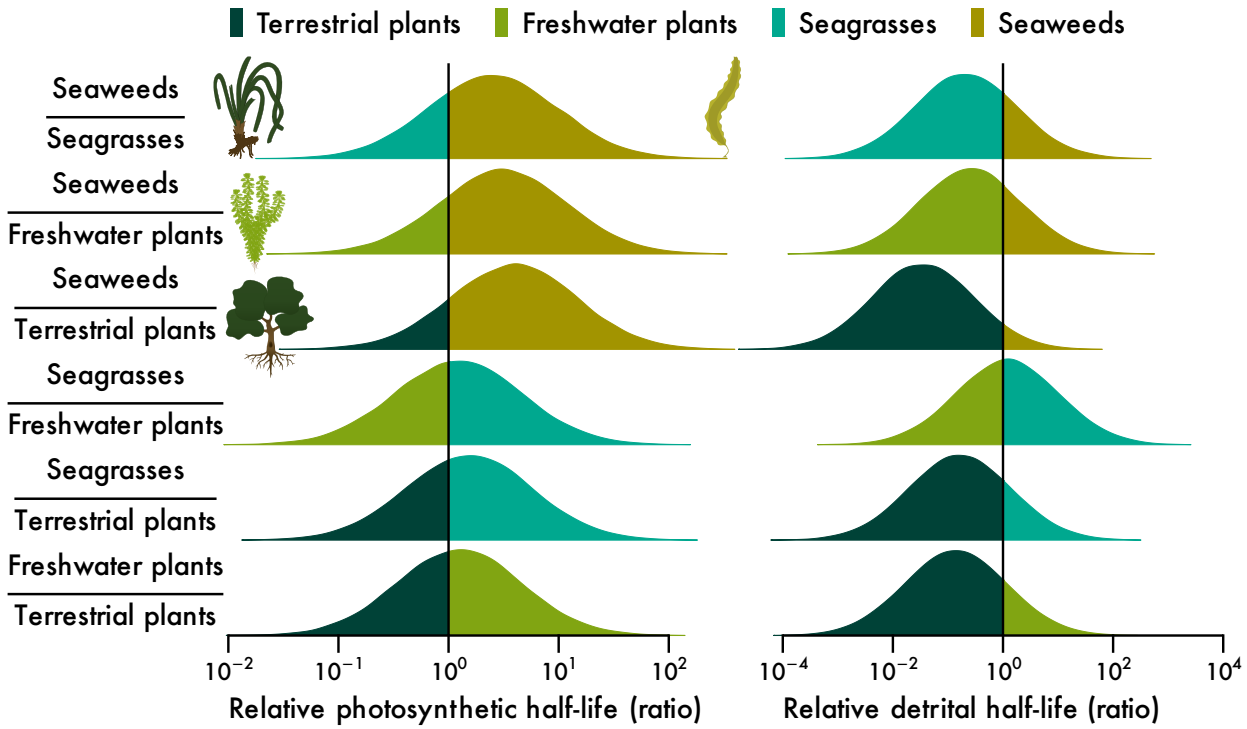


Figure S2. Photosynthetic and detrital half-life contrasts between plant groups. The group with more probability (coloured area) either side of the identity line at ratio = 1 has a greater half-life.

Supplementary tables

Table S1. Species included in the meta-analysis. Phyla are given in bold and orders are underlined. Seaweeds are comprised of Chlorophyta and Heterokontophyta. Seagrasses are marked with an asterisk. All remaining Alismatales are freshwater plants and the rest of Streptophyta is terrestrial. The longest experimental durations and largest numbers of studies, experiments and observations in each non-taxonomic group are highlighted in bold.

	Species	Days	Studies	Exper.	Observ.
Chlorophyta					
Ulvales					
Ulvaceae	*Ulva clathrata*	30	1	1	6
	Ulva intestinalis	30	1	1	6
	Ulva reticulata	30	1	1	6
Heterokontophyta					
Desmarestiales					
Desmarestiaceae	*Desmarestia anceps*	**313**	1	1	65
Fucales					
Durvillaeaceae	*Durvillaea antarctica*	14	2	16	448
Sargassaceae	*Sargassum fallax*	14	1	9	84
	Sargassum spinuligerum	14	1	13	159
Seirococcaceae	*Scytothalia dorycarpa*	14	1	6	48
Laminariales					
Laminariaceae	*Laminaria digitata*	46	4	6	191
	Laminaria hyperborea	168	**7**	**56**	**559**
	Laminaria ochroleuca	56	4	51	416
	Macrocystis pyrifera	55	4	23	375
	Saccharina latissima	39	2	2	55
Lessoniaceae	*Ecklonia radiata*	119	2	10	117
Tilopteridales					
Phyllariaceae	*Saccorhiza polyschides*	39	1	1	38
Streptophyta					
Alismatales					
Cymodoceaceae	*Amphibolis antarctica**	34	**1**	1	39
Hydrocharitaceae	*Elodea densa*	16	2	3	43
	*Halophila ovalis**	34	**1**	1	42
	Hydrilla verticillata	12	**7**	**16**	**506**
	*Thalassia testudinum**	52	**1**	1	14
	Vallisneria americana	**104**	1	1	5
	Vallisneria spiralis	9	6	6	59
Potamogetonaceae	*Potamogeton* sp.	**104**	1	1	5
	Stuckenia pectinata	9	5	5	55
Zosteraceae	*Zostera marina**	**85**	**1**	1	7
	*Zostera muelleri**	16	**1**	**4**	**58**
Apiales					
Apiaceae	*Coriandrum sativum*	9	1	1	24
	Petroselinum crispum	7	1	1	7
Asterales					
Asteraceae	*Chrysanthemum × morifolium*	12	1	2	30
	Helianthus annuus	0.024	1	1	36
	Helianthus tuberosus	0.026	1	1	36
	Inula racemosa	0.025	1	1	36

	Species	Days	Studies	Exper.	Observ.
	Smallanthus connatus	0.026	1	1	36
Brassicales					
Brassicaceae	*Arabidopsis thaliana*	6	2	3	60
	Brassica napus	1.7	1	25	125
	Brassica oleracea	10	1	3	22
	Nasturtium officinale	5	2	3	39
	Raphanus raphanistrum	6	1	4	20
	Sinapis alba	6	1	4	20
Tropaeolaceae	*Tropaeolum majus*	8	2	3	27
Caryophyllales					
Amaranthaceae	*Amaranthus cruentus*	0.026	1	1	36
	Celosia argentea	0.024	1	1	36
	Chenopodium berlandieri	0.27	1	2	26
	Gomphrena serrata	0.024	1	1	36
	Spinacia oleracea	10	3	4	85
Polygonaceae	*Koenigia weyrichii*	0.028	1	1	36
Cucurbitales					
Cucurbitaceae	*Cucumis sativus*	13	2	3	13
	Sicyos edulis	10	1	2	48
Ericales					
Theaceae	*Camellia sinensis*	4	1	2	11
Fabales					
Fabaceae	*Cercis canadensis*	0.0052	1	4	576
	Lathyrus oleraceus	17	2	2	42
	Phaseolus vulgaris	2.1	1	1	10
	Trifolium subterraneum	7	2	5	25
	Vigna unguiculata	4	1	10	170
Fagales					
Betulaceae	*Betula pendula*	0.029	1	1	36
Fagaceae	*Quercus alba*	0.01	1	1	24
	Quercus muehlenbergii	0.0052	1	1	192
	Quercus robur	0.024	2	2	228
	Quercus rubra	0.01	1	1	24
Juglandaceae	*Juglans nigra*	0.01	1	1	24
Lamiales					
Lamiaceae	*Salvia officinalis*	11	1	1	8
Laurales					
Lauraceae	*Cinnamomum tamala*	21	1	1	36
Pinales					
Pinaceae	*Larix sibirica*	0.025	1	1	36
	Pinus sibirica	0.023	1	1	33
	Pinus sylvestris	0.022	1	1	33
Poales					
Cyperaceae	*Bolboschoenus fluviatilis*	**104**	1	1	5
Poaceae	*Avena sativa*	8	8	28	161
	Hordeum vulgare	11	**13**	25	466
	Lolium multiflorum	5	1	3	44
	Lolium pratense	7	5	8	112
	Oryza sativa	27	11	**48**	327
	Panicum miliaceum	5	2	2	69
	Secale cereale	5	2	9	102

	Species	Days	Studies	Exper.	Observ.
	Triticum aestivum	10	**13**	41	646
	Triticum turgidum	6	1	1	7
	Triticum vulgare	6	1	2	48
	Zea mays	6	3	4	58
Typhaceae	*Typha angustifolia*	**104**	1	1	5
Polypodiales					
Nephrolepidaceae	*Nephrolepis exaltata*	10	1	4	24
Rosales					
Rosaceae	*Fragaria × ananassa*	0.0056	1	1	663
	Malus domestica	21	3	6	150
	Malus hupehensis	15	1	1	18
Sapindales					
Rutaceae	*Citrus × aurantium*	10	1	1	5
Sapindaceae	*Acer saccharinum*	0.25	1	4	36
	Acer truncatum	0.0052	1	1	192
Solanales					
Convolvulaceae	*Ipomoea batatas*	3	1	2	24
Solanaceae	*Nicotiana rustica*	10	1	2	180
	Nicotiana tabacum	9	1	2	24
	Solanum lycopersicum	5	1	6	192
	Solanum melongena	16	1	1	24
	Solanum tuberosum	0.026	1	1	36
Vitales					
Vitaceae	*Vitis vinifera*	0.0052	1	5	**1280**

Table S2. Predictions of photosynthetic half-life (days, μ in Equation 2) for new experiments on unobserved species in each plant group in this study as well as new plant groups. Estimates are stratified by observed differences in light condition and the type of response variable. All estimates are given as median (ln mean ± standard deviation).

Light	Response	Photosynthetic half-life
Terrestrial plants		
Light	Photosynthesis	13 (2.6 ± 0.94)
	Chlorophyll	18 (2.9 ± 0.93)
Dark	Photosynthesis	7.2 (2 ± 0.92)
	Chlorophyll	9.5 (2.3 ± 0.92)
Freshwater plants		
Light	Chlorophyll	19 (2.9 ± 0.98)
Dark	Photosynthesis	11 (2.4 ± 0.95)
	Chlorophyll	16 (2.8 ± 0.95)
Seagrasses		
Light	Photosynthesis	19 (2.9 ± 0.98)
	Chlorophyll	26 (3.3 ± 0.98)
Dark	Chlorophyll	16 (2.8 ± 1)
Seaweeds		
Light	Photosynthesis	51 (3.9 ± 0.94)
	Chlorophyll	81 (4.4 ± 0.96)
Dark	Photosynthesis	24 (3.2 ± 1)
All plants		
Light	Photosynthesis	17 (2.8 ± 1.1)
	Chlorophyll	24 (3.2 ± 1.1)
Dark	Photosynthesis	12 (2.4 ± 1.1)
	Chlorophyll	16 (2.8 ± 1.1)

Table S3. Predictions of photosynthetic half-life (days, μ in Equation 2) for new experiments on each species in this study (cf. Table S1 for complete taxonomy). Estimates are stratified by observed differences in light condition and the type of response variable. All estimates are given as median (ln mean ± standard deviation).

Species	Light	Response	Photosynthetic half-life
Seaweeds			
Desmarestia anceps	Light	Chlorophyll	154 (5 ± 0.9)
Durvillaea antarctica	Light	Photosynthesis	59 (4.1 ± 0.87)
		Chlorophyll	95 (4.6 ± 0.9)
Ecklonia radiata	Light	Photosynthesis	32 (3.5 ± 0.78)
	Dark	Photosynthesis	15 (2.7 ± 0.87)
Laminaria digitata	Light	Photosynthesis	68 (4.2 ± 0.86)
		Chlorophyll	109 (4.7 ± 0.88)
Laminaria hyperborea	Light	Photosynthesis	62 (4.1 ± 0.73)
		Chlorophyll	99 (4.6 ± 0.79)
Laminaria ochroleuca	Light	Photosynthesis	68 (4.2 ± 0.74)
		Chlorophyll	108 (4.7 ± 0.79)
Macrocystis pyrifera	Light	Photosynthesis	60 (4.1 ± 0.84)
		Chlorophyll	96 (4.6 ± 0.86)
Saccharina latissima	Light	Photosynthesis	60 (4.1 ± 0.9)
		Chlorophyll	96 (4.6 ± 0.91)
Saccorhiza polyschides	Light	Chlorophyll	87 (4.5 ± 0.92)

Species	Light	Response	Photosynthetic half-life
Sargassum fallax	Light	Photosynthesis	30 (3.4 ± 0.82)
Sargassum spinuligerum	Light	Photosynthesis	64 (4.2 ± 0.86)
		Chlorophyll	102 (4.6 ± 0.89)
Scytothalia dorycarpa	Light	Photosynthesis	45 (3.8 ± 0.95)
Ulva clathrata	Light	Chlorophyll	85 (4.4 ± 0.94)
Ulva intestinalis	Light	Chlorophyll	84 (4.4 ± 0.94)
Ulva reticulata	Light	Chlorophyll	85 (4.4 ± 0.94)
Seagrasses			
Amphibolis antarctica	Light	Photosynthesis	14 (2.6 ± 0.88)
Halophila ovalis	Light	Photosynthesis	27 (3.3 ± 0.91)
Thalassia testudinum	Light	Chlorophyll	26 (3.2 ± 0.87)
Zostera marina	Light	Chlorophyll	26 (3.3 ± 0.87)
Zostera muelleri	Light	Chlorophyll	25 (3.2 ± 0.86)
	Dark	Chlorophyll	16 (2.8 ± 0.86)
Freshwater plants			
Elodea densa	Light	Chlorophyll	18 (2.9 ± 0.89)
	Dark	Chlorophyll	15 (2.7 ± 0.86)
Hydrilla verticillata	Light	Chlorophyll	14 (2.6 ± 0.79)
	Dark	Photosynthesis	7.7 (2 ± 0.78)
	Dark	Chlorophyll	12 (2.5 ± 0.78)
Potamogeton sp.	Dark	Chlorophyll	25 (3.2 ± 0.88)
Stuckenia pectinata	Dark	Photosynthesis	7.1 (2 ± 0.8)
	Dark	Chlorophyll	11 (2.4 ± 0.82)
Vallisneria americana	Dark	Chlorophyll	25 (3.2 ± 0.88)
Vallisneria spiralis	Dark	Photosynthesis	7.8 (2.1 ± 0.79)
	Dark	Chlorophyll	12 (2.5 ± 0.81)
Terrestrial plants			
Acer saccharinum	Light	Photosynthesis	13 (2.6 ± 0.93)
Acer truncatum	Light	Photosynthesis	13 (2.6 ± 0.94)
Amaranthus cruentus	Light	Photosynthesis	13 (2.6 ± 0.93)
Arabidopsis thaliana	Dark	Photosynthesis	4.3 (1.5 ± 0.81)
	Dark	Chlorophyll	5.8 (1.8 ± 0.8)
Avena sativa	Light	Chlorophyll	11 (2.4 ± 0.78)
	Dark	Chlorophyll	5.8 (1.8 ± 0.76)
Betula pendula	Light	Photosynthesis	13 (2.6 ± 0.94)
Bolboschoenus fluviatilis	Dark	Chlorophyll	20 (3 ± 0.9)
Brassica napus	Light	Chlorophyll	18 (2.9 ± 0.91)
Brassica oleracea	Dark	Chlorophyll	8.9 (2.2 ± 0.85)
Camellia sinensis	Light	Photosynthesis	14 (2.7 ± 0.91)
Celosia argentea	Light	Photosynthesis	13 (2.6 ± 0.93)
Cercis canadensis	Light	Photosynthesis	13 (2.6 ± 0.93)
Chenopodium berlandieri	Light	Photosynthesis	13 (2.6 ± 0.94)
Chrysanthemum × morifolium	Dark	Chlorophyll	9.4 (2.2 ± 0.82)
Cinnamomum tamala	Dark	Chlorophyll	9.6 (2.3 ± 0.84)
Citrus × aurantium	Light	Chlorophyll	19 (2.9 ± 0.9)
Coriandrum sativum	Dark	Chlorophyll	9.6 (2.3 ± 0.87)
Cucumis sativus	Light	Chlorophyll	19 (3 ± 0.89)
	Dark	Chlorophyll	10 (2.3 ± 0.88)
Fragaria × ananassa	Light	Photosynthesis	13 (2.6 ± 0.93)
Gomphrena serrata	Light	Photosynthesis	13 (2.6 ± 0.93)
Helianthus annuus	Light	Photosynthesis	13 (2.6 ± 0.94)

Species	Light	Response	Photosynthetic half-life
Helianthus tuberosus	Light	Photosynthesis	13 (2.6 ± 0.93)
Hordeum vulgare	Light	Photosynthesis	9.2 (2.2 ± 0.77)
		Chlorophyll	12 (2.5 ± 0.76)
	Dark	Photosynthesis	5 (1.6 ± 0.75)
		Chlorophyll	6.6 (1.9 ± 0.75)
Inula racemosa	Light	Photosynthesis	13 (2.6 ± 0.94)
Ipomoea batatas	Dark	Photosynthesis	7.9 (2.1 ± 0.9)
		Chlorophyll	11 (2.4 ± 0.9)
Juglans nigra	Light	Photosynthesis	13 (2.6 ± 0.93)
Koenigia weyrichii	Light	Photosynthesis	13 (2.6 ± 0.93)
Larix sibirica	Light	Photosynthesis	13 (2.6 ± 0.94)
Lathyrus oleraceus	Light	Photosynthesis	16 (2.8 ± 0.89)
		Chlorophyll	22 (3.1 ± 0.88)
Lolium multiflorum	Light	Photosynthesis	12 (2.5 ± 0.92)
		Chlorophyll	16 (2.8 ± 0.92)
Lolium pratense	Dark	Photosynthesis	4.1 (1.4 ± 0.77)
		Chlorophyll	5.5 (1.7 ± 0.77)
Malus domestica	Light	Photosynthesis	24 (3.2 ± 0.86)
		Chlorophyll	33 (3.5 ± 0.85)
	Dark	Photosynthesis	13 (2.6 ± 0.85)
		Chlorophyll	18 (2.9 ± 0.85)
Malus hupehensis	Dark	Chlorophyll	13 (2.6 ± 0.88)
Nasturtium officinale	Dark	Chlorophyll	4.3 (1.5 ± 0.81)
Nephrolepis exaltata	Light	Photosynthesis	18 (2.9 ± 0.87)
		Chlorophyll	24 (3.2 ± 0.86)
	Dark	Photosynthesis	9.8 (2.3 ± 0.86)
		Chlorophyll	13 (2.6 ± 0.85)
Nicotiana rustica	Dark	Photosynthesis	8.5 (2.1 ± 0.84)
		Chlorophyll	11 (2.4 ± 0.83)
Nicotiana tabacum	Light	Photosynthesis	16 (2.8 ± 0.89)
		Chlorophyll	21 (3.1 ± 0.88)
Oryza sativa	Light	Photosynthesis	10 (2.4 ± 0.77)
		Chlorophyll	14 (2.6 ± 0.76)
	Dark	Photosynthesis	5.7 (1.7 ± 0.76)
		Chlorophyll	7.5 (2 ± 0.75)
Panicum miliaceum	Dark	Chlorophyll	5.4 (1.7 ± 0.82)
Petroselinum crispum	Dark	Chlorophyll	10 (2.3 ± 0.89)
Phaseolus vulgaris	Light	Photosynthesis	13 (2.6 ± 0.93)
Pinus sibirica	Light	Photosynthesis	13 (2.6 ± 0.93)
Pinus sylvestris	Light	Photosynthesis	13 (2.6 ± 0.94)
Quercus alba	Light	Photosynthesis	13 (2.6 ± 0.94)
Quercus muehlenbergii	Light	Photosynthesis	13 (2.6 ± 0.93)
Quercus robur	Light	Photosynthesis	13 (2.6 ± 0.93)
Quercus rubra	Light	Photosynthesis	13 (2.6 ± 0.94)
Raphanus raphanistrum	Light	Photosynthesis	14 (2.7 ± 0.89)
		Chlorophyll	19 (3 ± 0.89)
Salvia officinalis	Dark	Chlorophyll	12 (2.5 ± 0.89)
Secale cereale	Light	Photosynthesis	13 (2.6 ± 0.86)
		Chlorophyll	18 (2.9 ± 0.85)
	Dark	Photosynthesis	7.2 (2 ± 0.85)
		Chlorophyll	9.5 (2.3 ± 0.85)

Species	Light	Response	Photosynthetic half-life
Sicyos edulis	Dark	Photosynthesis	7.4 (2 ± 0.82)
		Chlorophyll	9.9 (2.3 ± 0.82)
Sinapis alba	Light	Photosynthesis	15 (2.7 ± 0.89)
		Chlorophyll	20 (3 ± 0.89)
Smallanthus connatus	Light	Photosynthesis	13 (2.6 ± 0.93)
Solanum lycopersicum	Light	Photosynthesis	11 (2.4 ± 0.85)
Solanum melongena	Light	Photosynthesis	15 (2.7 ± 0.88)
Solanum tuberosum	Light	Photosynthesis	13 (2.6 ± 0.93)
Spinacia oleracea	Dark	Photosynthesis	5.8 (1.8 ± 0.8)
		Chlorophyll	7.7 (2 ± 0.79)
Trifolium subterraneum	Light	Photosynthesis	12 (2.5 ± 0.88)
		Chlorophyll	16 (2.8 ± 0.88)
Triticum aestivum	Light	Photosynthesis	5.5 (1.7 ± 0.75)
		Chlorophyll	7.3 (2 ± 0.75)
	Dark	Photosynthesis	3 (1.1 ± 0.74)
		Chlorophyll	4 (1.4 ± 0.74)
Triticum turgidum	Light	Chlorophyll	18 (2.9 ± 0.92)
Triticum vulgare	Light	Photosynthesis	13 (2.6 ± 0.88)
	Dark	Photosynthesis	7.3 (2 ± 0.87)
Tropaeolum majus	Light	Chlorophyll	20 (3 ± 0.87)
	Dark	Chlorophyll	11 (2.4 ± 0.87)
Typha angustifolia	Dark	Chlorophyll	22 (3.1 ± 0.91)
Vigna unguiculata	Dark	Photosynthesis	5.3 (1.7 ± 0.78)
		Chlorophyll	7.2 (2 ± 0.78)
Vitis vinifera	Light	Photosynthesis	13 (2.6 ± 0.93)
Zea mays	Light	Photosynthesis	14 (2.6 ± 0.91)
		Chlorophyll	19 (2.9 ± 0.9)
	Dark	Chlorophyll	10 (2.3 ± 0.89)

Supplementary references

Studies included in the meta-analysis in addition to the present study. Note that 130 references are listed because the data from three studies were not independent and thus merged.

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